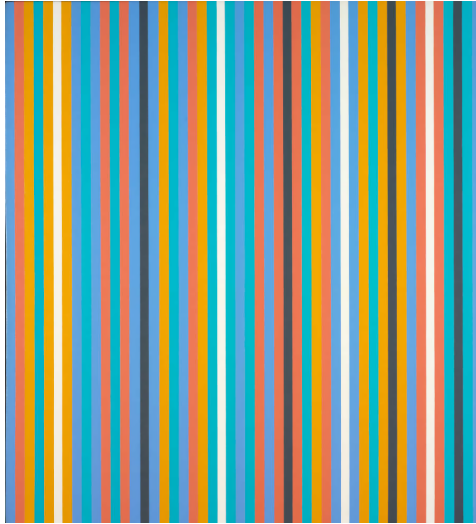




Bridget Riley (1982) Gather

The Visual System

Retina

Computational Neuroscience

University of Bristol

mrule

Learning outcomes:

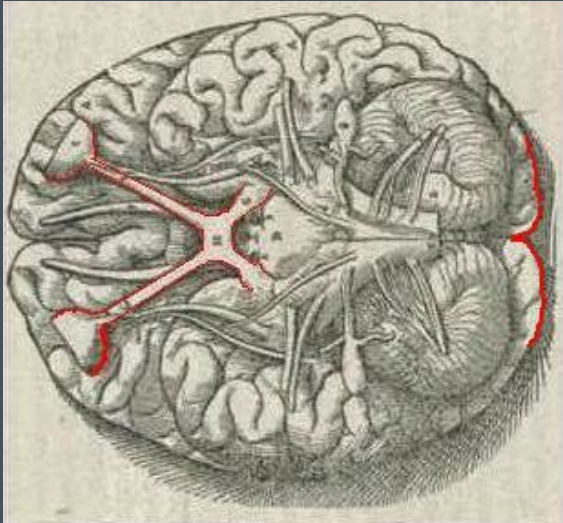
- ▶ What types of visual processing occurs in the retina?
- ▶ What features are Retinal Ganglion Cells (RGCs) sensitive to?
- ▶ How do RGCs compress visual information?

What you need to take away for the assessments:

- ▶ Be able to answer fact-based questions about topographic maps and sparse coding
- ▶ Know how visual features are transformed as we move deeper into the visual system.
 - What feature(s) are the neuron sensitive to? (what is it's tuning?)
 - Where in the eye/brain do you find neurons with this tuning?
 - What is the name of this neuron type?
 - How would you describe its tuning mathematically?

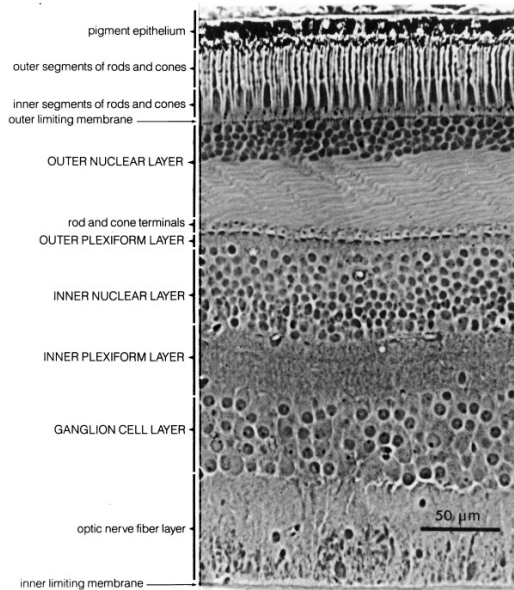
Potentially helpful for Bard's talk:

- ▶ Retinotopic transformation
- ▶ V1 cortical microcircuit and recurrent (lateral) connections



Andreas Vesalius (1543) De Humani Corporis Fabrica (Wikimedia)

Retina



Dowling and Boycott 1966

The retina processes and compresses visual information to transmit via the optic nerve. Its output neurons (Retinal Ganglion Cells, RGCs) are sensitive to contrasts in time, space (less so absolute luminance)

The retina contains lateral connections to implement surround inhibition

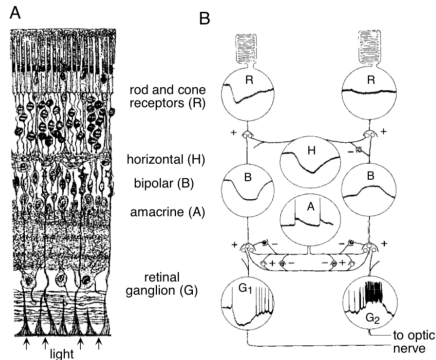
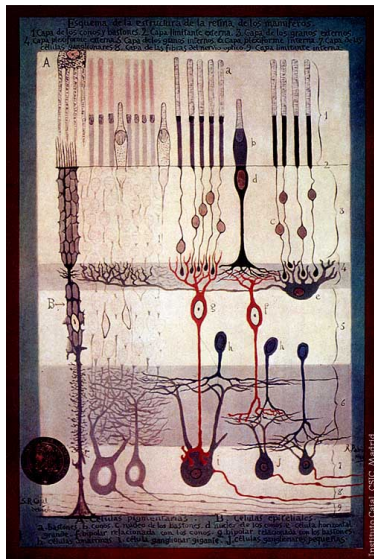


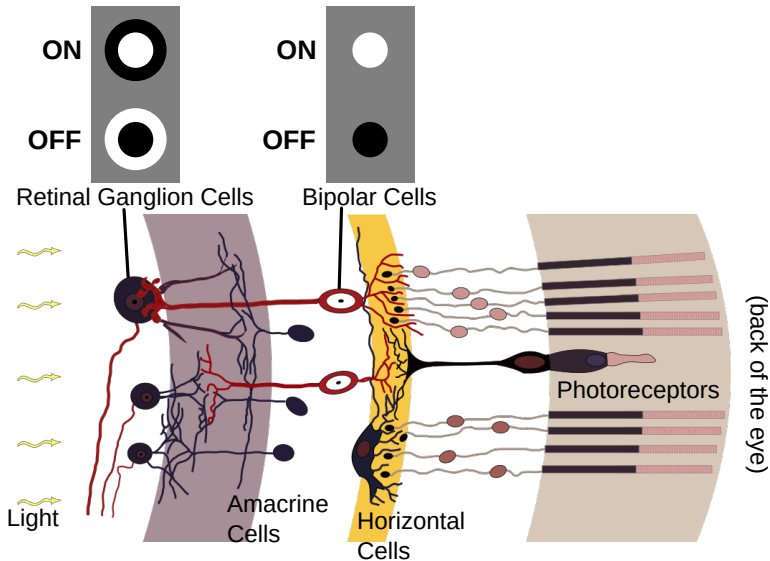
Figure 2.4: A) An anatomical diagram of the circuitry of the retina of a dog. Cell types are identified at right. In the intact eye, illumination is, counter-intuitively, from the bottom of this figure. B) Intracellular recordings from retinal neurons of the mudpuppy responding to flash of light lasting for one second. In the column of cells on the left side of the diagram, the resulting hyperpolarizations are about 4 mV in the rod and retinal ganglion cells, and 8 mV in the bipolar cell. Plus and minus represent excitatory and inhibitory synapses respectively. (A adapted from Nicholls et al., 1992; drawing from Cajal, 1911. B data from Werblin and Dowling 1969; figure adapted from Dowling, 1992.)

The retina contains lateral connections to implement surround inhibition



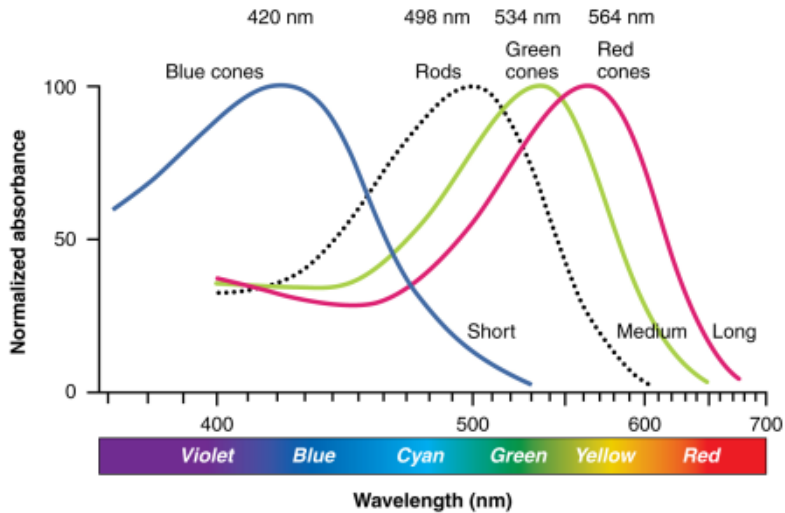
Ramón y Cajal, circa 1900

Retinal Ganglion cells show centre-surround spatial tuning

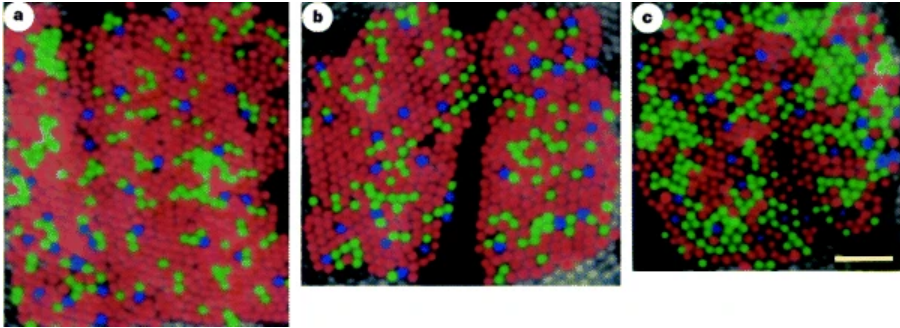


Ramón y Cajal (1911) modified from derivative work by A. Friedrich (wikimedia commons)

Most of you probably have 2–4 distinct types of cones in your retina sensitive to different color wavelengths (3 non-anomalous cones, “trichomat” being the most common). In trichomat vision, red–green and yellow–blue contrast information are also transmitted via centre-surround opponent channels.

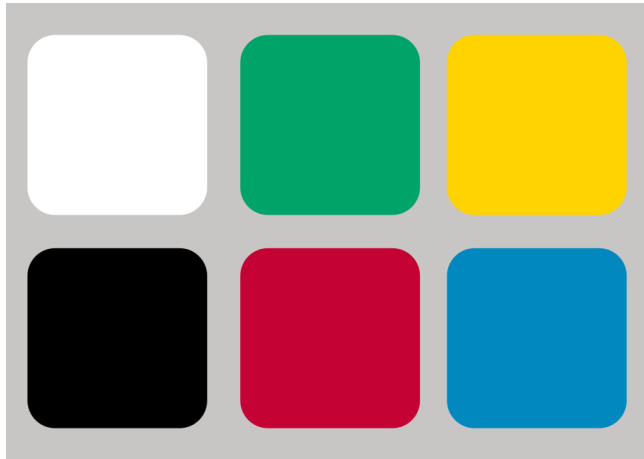


Cone sensitivity, [OpenStax College](#)

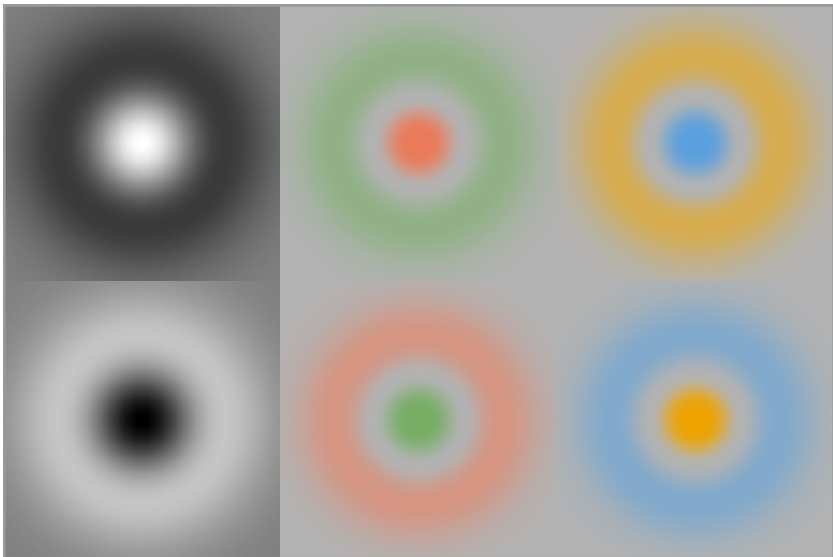


Roorda and Williams, Nature (1999)

Opponent channels



These are roughly the three color channels Retinal Ganglion Cells (RGCs) use to transmit color information in people with trichomat vision. The actual contrasts are processed a bit in the LGN relay and the Thalamus, and RGC tuning is a bit more diverse than this (and includes e.g. temporal contrasts).



The Visual System: Retina

How would you describe the tuning of a RGC? _____

Where can I find neurons tuned to a dark spot on a bright background (or vice versa)? _____

Do photoreceptors show centre-surround tuning? ____

Bright/dark, red/green, yellow/blue approximate the coding scheme for retinal _____
in non-anomalous trichomat vision.

The Visual System: Retina

How would you describe the tuning of a RGC? centre-surround

Where can I find neurons tuned to a dark spot on a bright background (or vice versa)? retina

Do photoreceptors show centre-surround tuning? no

Bright/dark, red/green, yellow/blue approximate the coding scheme for retinal opponent color coding in non-anomalous trichomat vision.

Just for fun, try playing this later:

<https://eyewire.org/explore>

end

